

Section 3

Myerson's Lemma

Single-Parameter Environments

- Some number n of agents.
 - Each agent i has a private nonnegative valuation v_i
 - her value “per unit of stuff” that she acquires.
- There is a feasible set X .
 - Each element of X is a nonnegative n vector $(x_1, x_2, \dots, x_n)$
 - where x_i denotes the “amount of stuff” given to agent i .

Examples

Example 3.1 (Single-Item Auction) In a single-item auction (Section 2.1), X is the set of 0-1 vectors that have at most one 1—that is, $\sum_{i=1}^n x_i \leq 1$.

Example 3.2 (k -Unit Auction) With k identical items and the constraint that each bidder gets at most one (Exercise 2.3), the feasible set is the set of 0-1 vectors that satisfy $\sum_{i=1}^n x_i \leq k$.

Examples

Example 3.3 (Sponsored Search Auction) In a sponsored search auction (Section 2.6), X is the set of n -vectors corresponding to assignments of bidders to slots, where each slot is assigned to at most one bidder and each bidder is assigned to at most one slot. If bidder i is assigned to slot j , then the component x_i equals the click-through rate α_j of her slot.

Example 3.4 (Public Project) Deciding whether or not to build a public project that can be used by all, such as a new bridge, can be modeled by the set $X = \{(0, 0, \dots, 0), (1, 1, \dots, 1)\}$.

Allocation and Payment Rules

1. Collect bids $b = (b_1, \dots, b_n)$ from all agents. The vector b is called the bid vector or bid profile.
2. [**allocation rule**] Choose a feasible allocation $x(b) \in X \subseteq R^n$ as a function of the bids.
3. [**payment rule**] Choose payments $p(b) \in R^n$ as a function of the bids.

direct-revelation mechanisms vs

indirect mechanism (iterative ascending auction, exe. 2.7)

quasilinear utility model

In a mechanism with allocation and payment rules x and p agent i receives utility

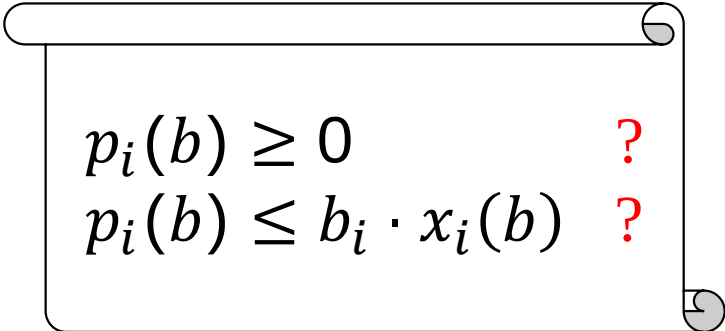
$$u_i(b) = v_i \cdot x_i(b) - p_i(b)$$

when the bid profile is b .

We focus on payment rules that satisfy

$$p_i(b) \in [0, b_i \cdot x_i(b)]$$

for every agent i and bid profile b .


$$p_i(b) \geq 0 \quad ?$$

$$p_i(b) \leq b_i \cdot x_i(b) \quad ?$$

Statement of Myerson's Lemma

Definition 3.5 (*Implementable Allocation Rule*) An allocation rule x for a single-parameter environment is implementable if there is a payment rule p such that the direct-revelation mechanism (x, p) is **DSIC**.

Definition 3.6 (*Monotone Allocation Rule*) An allocation rule x for a single-parameter environment is monotone if for every agent i and bids b_{-i} by the other agents, the allocation $x_i(z, b_{-i})$ to i is **nondecreasing** in her bid z .

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bidding higher can only
get you more stuff.

Examples

For example, the **single-item auction allocation rule** that awards the item to the **highest bidder** is monotone: if you're the winner and you raise your bid (keeping other bids fixed), you continue to win.

By contrast, awarding the item to the **second-highest bidder** is a nonmonotone allocation rule: if you're the winner and you raise your bid high enough, you lose.

The **welfare-maximizing allocation rule** for sponsored search auctions (Example 3.3), with the **i -th highest bidder** awarded the **i -th best slot**, is monotone.

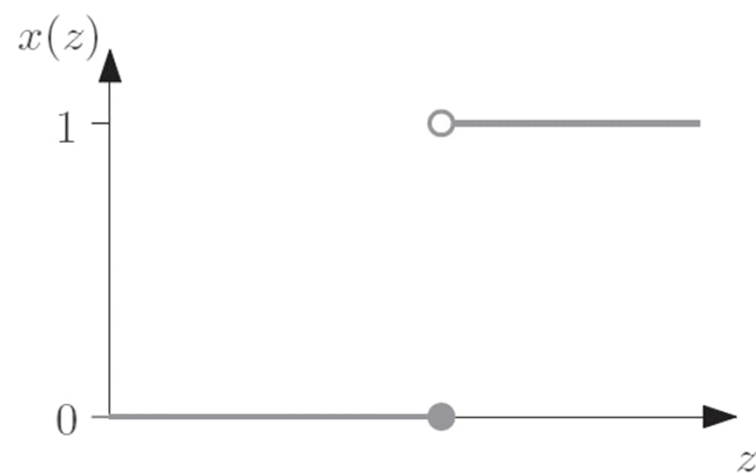
Myerson's Lemma

Theorem 3.7 (*Myerson's Lemma*) Fix a single-parameter environment.

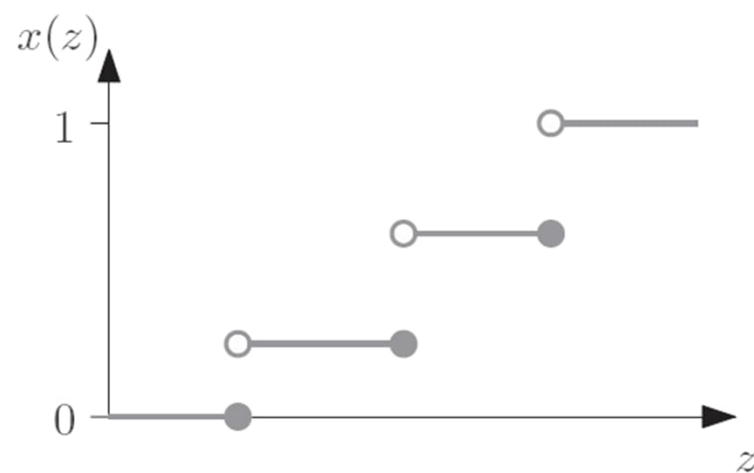
- (a) An allocation rule x is **implementable** if and only if it is **monotone**.
- (b) If x is **monotone**, then there is a **unique payment rule** for which the direct-revelation mechanism (x, p) is **DSIC** and $p_i(b) = 0$ whenever $b_i = 0$.
- (c) The payment rule in (b) is given by **an explicit formula**.

Myerson's lemma is the foundation on which we'll build most of our mechanism design theory.

Myerson's Lemma



(a) 0-1 monotone curve



(b) Piecewise constant monotone curve

Figure 3.1: Examples of allocation curves $x(\cdot)$.

Proof by Picture

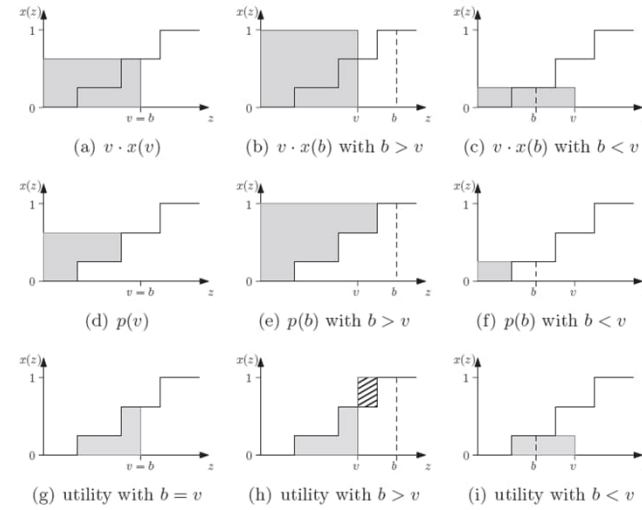


Figure 3.2: Proof by picture that the payment rule in (3.5), coupled with the given monotone and piecewise constant allocation rule, yields a DSIC mechanism. The three columns consider the cases of truthful bidding, overbidding, and underbidding, respectively. The three rows show the welfare $v \cdot x(b)$, the payment $p(b)$, and the utility $v \cdot x(b) - p(b)$, respectively. In (h), the solid region represents positive utility and the lined region represents negative utility.